



Harshavarthan Venkatesan

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● ABOUT ME

Artificial Intelligence and Data Science undergraduate with hands-on experience in deep learning, computer vision, embedded systems and full-stack development. Currently Student Lead for Technology and Research at Microcosm, leading a Harvard-funded computer vision project on elephant behaviour and human-elephant conflict mitigation. Former embedded engineer with Team Sea Sakthi at the Monaco Energy Boat Challenge. Interested in applying AI to real-world problems.

● EDUCATION & TRAINING

09/09/2023 - Current - COIMBATORE, INDIA

BACHELOR OF TECHNOLOGY IN ARTIFICIAL INTELLIGENCE AND DATA SCIENCE- KUMARAGURU COLLEGE OF TECHNOLOGY

AI & Machine Learning, Applied Machine Learning, Neural Networks & Deep Learning, Reinforcement Learning, Data Science, Data Mining & Modelling, EDA & Visualisation, Cloud Architecture, Industrial IoT, Business Intelligence.

Field(s) of study: Artificial Intelligence and Data Science | **Final grade:** 7.36/10 | **Level in EQF:** 6 | **Number of credits:** 162 | **Thesis:** Elephant Behavioural Analysis and Mitigation System – AI-Powered Computer Vision for Non-Invasive Elephant Behaviour Monitoring and Human–Elephant Conflict Mitigation (Supported by a USD 11,000 Harvard University grant)

01/01/2026 - 30/06/2026 - COIMBATORE, INDIA

GRADUATE CERTIFICATE IN INDUSTRIAL DIGITAL TECHNOLOGIES- FORGE INNOVATION & VENTURES

Natural Language Processing, Computer Vision Techniques, Startup Fundamentals, Applied Design Thinking, Prototype Development, Industrial IoT

Final grade: Semester VI GPA – 9.13/10

01/06/2022 - 05/04/2023 - COIMBATORE, INDIA

HIGHER SECONDARY CERTIFICATE (HSC) – TAMIL NADU STATE BOARD- LISIEUX MATRICULATION HIGHER SECONDARY SCHOOL

Computer Science, Physics, Chemistry, Mathematics, English, Tamil

Field(s) of study: Computer Science | **Final grade:** 91%

01/06/2020 - 30/04/2021 - COIMBATORE, INDIA

SECONDARY SCHOOL LEAVING CERTIFICATE (SSLC) – TAMIL NADU STATE BOARD- LISIEUX MATRICULATION HIGHER SECONDARY SCHOOL

Mathematics, Science, Social Science, English, Tamil

● WORK EXPERIENCE

01/07/2026 - CURRENT - COIMBATORE, INDIA

STUDENT LEAD – TECHNOLOGY AND RESEARCH PROJECTSMICROCOSM, KUMARAGURU COLLEGE OF TECHNOLOGY - (INTERNSHIP)

Lead, mentor and contribute hands-on to student research projects, including the Harvard-funded elephant, diary-based computer vision, campus carbon-footprint and dashboard projects.

Website: <https://microcosm.kct.ac.in/>

01/10/2024 - 30/09/2025 - COIMBATORE, INDIA

EMBEDDED ENGINEERTeam SEA SAKTHI, KUMARAGURU COLLEGE OF TECHNOLOGY - (INTERNSHIP)

Contributed to the embedded systems team developing the Yali 4.0 electric catamaran for the 12th Monaco Energy Boat Challenge and an autonomous river-cleaning boat, working on cloud telemetry, FPV-AR and real-time monitoring, and independently built the team website.

Website: <https://seasakthi.com/>

● PROJECTS

01/10/2024 - 30/09/2025

Yali 4.0 - TeamSeaSakthi Contributed to the embedded systems of an electric catamaran and an autonomous river-cleaning boat, working with microcontrollers, sensors, communication protocols, FPV-AR, cloud telemetry and dashboards, and represented the team at the 12th Monaco Energy Boat Challenge (8th place, Communication and Townhall Cup).

01/05/2026 - 30/04/2027

Elephant Behavioural Analysis and Mitigation System Lead a 13-member team on a USD 11,000 Harvard Mittal Institute grant project, contributing to models for elephant classification, identification, herd, feeding, gait and aggression analysis, and conflict-risk detection using YOLO, ByteTrack, CNN-XGBoost, MobileNetV3, DeepLabCut and 3D ResNet-18, integrated into an AI monitoring dashboard.

01/06/2026 - 06/10/2026

Sustainability Intelligence Dashboard - Kumaraguru Institutions Co-developed a campus sustainability platform (Python, FastAPI, PostgreSQL) as lead of a 4-member team, covering energy, GHG, water and waste data with Scope 1 and 2 GHG calculations aligned with ISO 14064-1:2018.

Link <https://sustainability.kct.ac.in>

17/07/2026 - 17/10/2026

K-geo portal Led a 3-member team and built a Next.js and Supabase portal featuring Gemini-powered university recommendations, role-based access control, and automated PDF reports.

10/07/2026 - CURRENT

Sugarcane Research Analytical Platform Develop and lead a full-stack React and Supabase analytics platform for a Microcosm and Environmental Defense Fund field study (3 plots, 14 treatments), featuring growth, fertiliser-response and heatmap analytics.

● SKILLS

computer vision | deep learning | machine learning | embedded systems | Internet of Things | Python (computer programming) | JavaScript | Embedded C | TypeScript | CSS | SQL | PostgreSQL | Full-Stack Development | ICT communications protocols | ICT system integration | sensors | perform data analysis | project management | manage teamwork | communicate with stakeholders | think innovatively | think analytically | evaluate project plans

● LANGUAGE SKILLS

Mother tongue(s): **TAMIL**

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C1	B2	B2	B2
GERMAN	A2	A2	A2	A2	A2
HINDI	B1	B1	B1	B1	B1

● HONOURS AND AWARDS

05/07/2025 Yacht Club de Monaco

Certificate of Achievement – Energy Class Team Sea Sakthi, Kumaraguru College of Technology – 8th Place and Communication and Townhall cup at the 12th Monaco Energy Boat Challenge (1–5 July 2025).

Link https://harshavarthan-umber.vercel.app/MEBC_Certificate.pdf

15/08/2025 Kumaraguru Institutions

Certificate of Recognition Recognition for contributions to Team Sea Sakthi, which represented Kumaraguru College of Technology at the Monaco Energy Boat Challenge 2025.

Link https://harshavarthan-umber.vercel.app/KCT_Certificate.pdf

29/09/2026 Microcosm

Appreciation letter Appreciated for taking initiative and leading technical and research projects at Microcosm, demonstrating dedication through mentoring, team upscaling, innovation, and a strong commitment to driving impactful outcomes.

Link https://harshavarthan-umber.vercel.app/Appreciation_Letter_Microcosm.pdf

17/09/2026 Kumaraguru Global Engagement Office

Appreciation letter Recognized for taking the initiative and leading the development of a portal to track students pursuing opportunities abroad. Appreciated for demonstrating responsibility, ownership, and innovation in supporting the Global Engagement Office.

Link https://harshavarthan-umber.vercel.app/Appreciation_Letter_Kgeo.pdf

● CERTIFICATIONS

15/07/2026 FORGE Innovation & Ventures

Graduate Certificate in Industrial Digital Technologies – PROTOSEM Innovation Engineers

Project Based - Completed the PROTOSEM Innovation Engineers programme (Cohort 1.17), applying technical knowledge through hands-on project development, prototyping, and innovation to solve real-world industrial problems.

04/09/2026 DeepLearning.AI

Deep Learning Specialization

Online - Completed the Deep Learning Specialization covering neural networks, CNNs, sequence models, and practical deep learning applications.

17/03/2025 University of Illinois Urbana-Champaign

Data Visualization

Online - Completed a course covering data visualization techniques for analyzing and communicating data effectively using visual methods.